

Chapter 2

Double and Depth Buffering

Interactive Computer Graphics

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Outline

- Single Buffering
- Double Buffering
- Depth Buffering

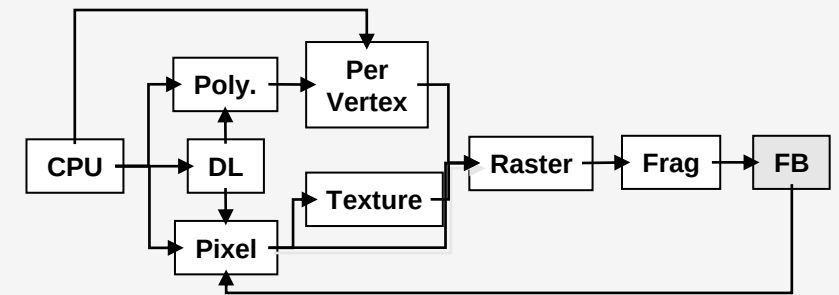
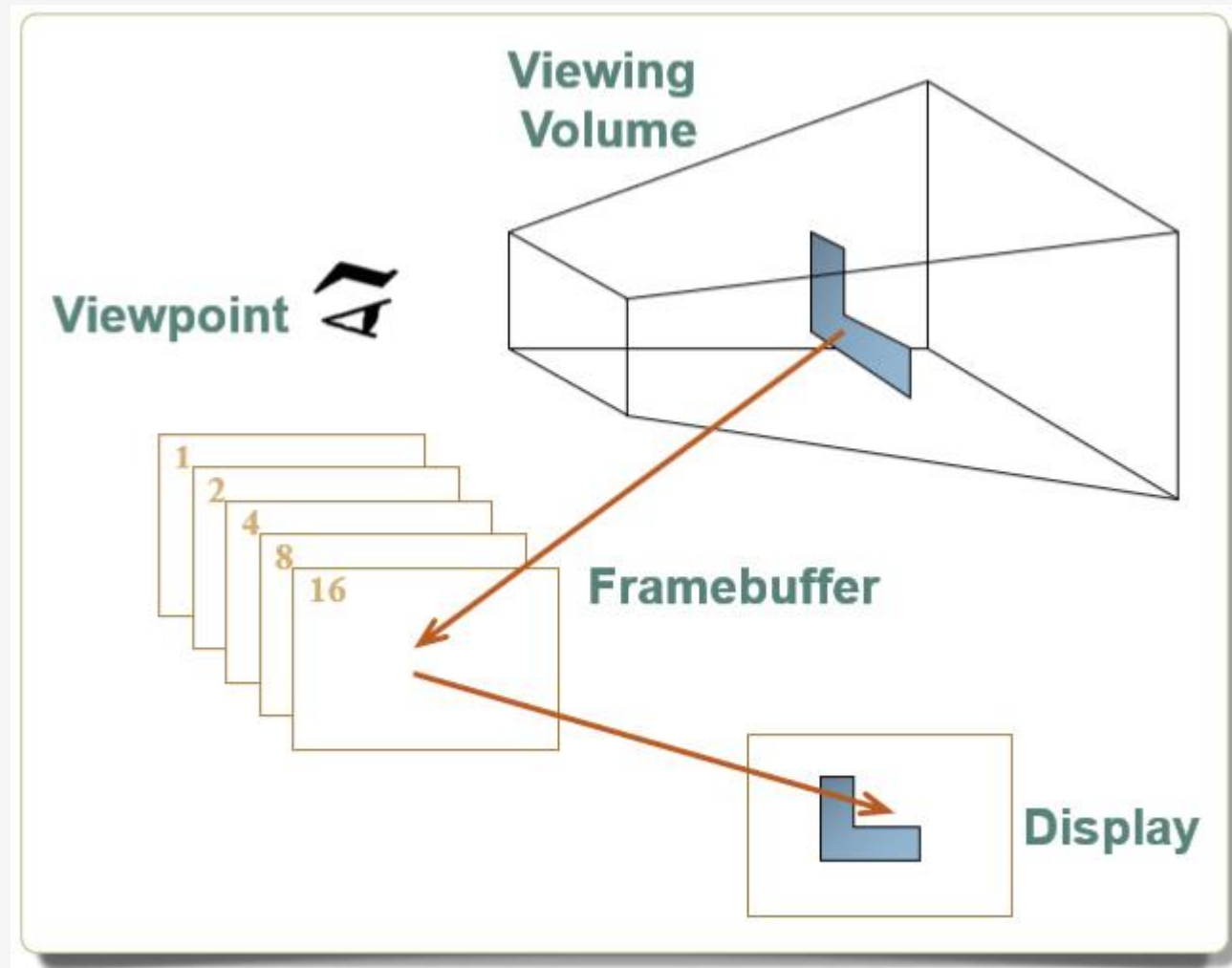
Flickering on Display with Single Framebuffer



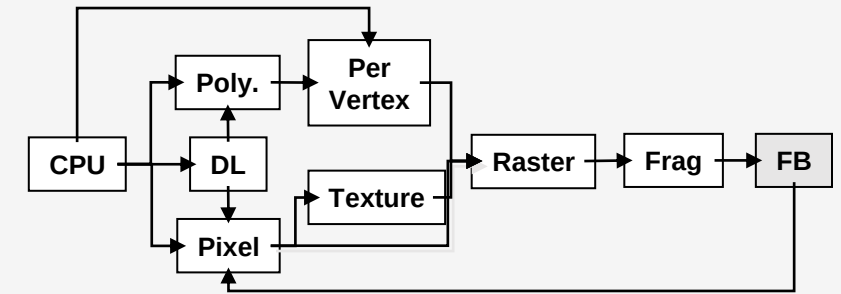
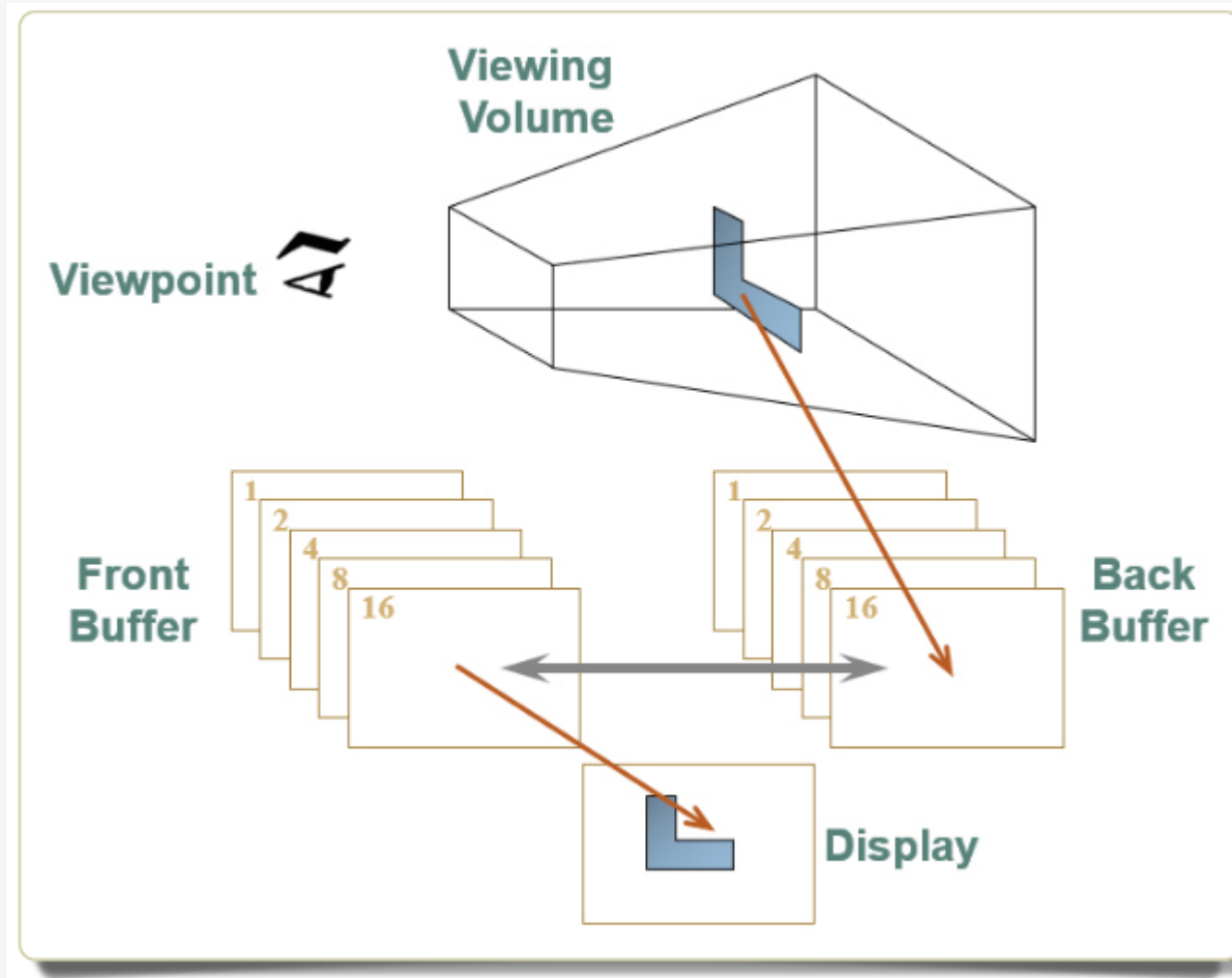
Source:

<https://renesasrulz.com/synergy/f/synergy---forum/14962/flickering-on-display-with-single-frame-buffer>

Single Buffering



Double Buffering



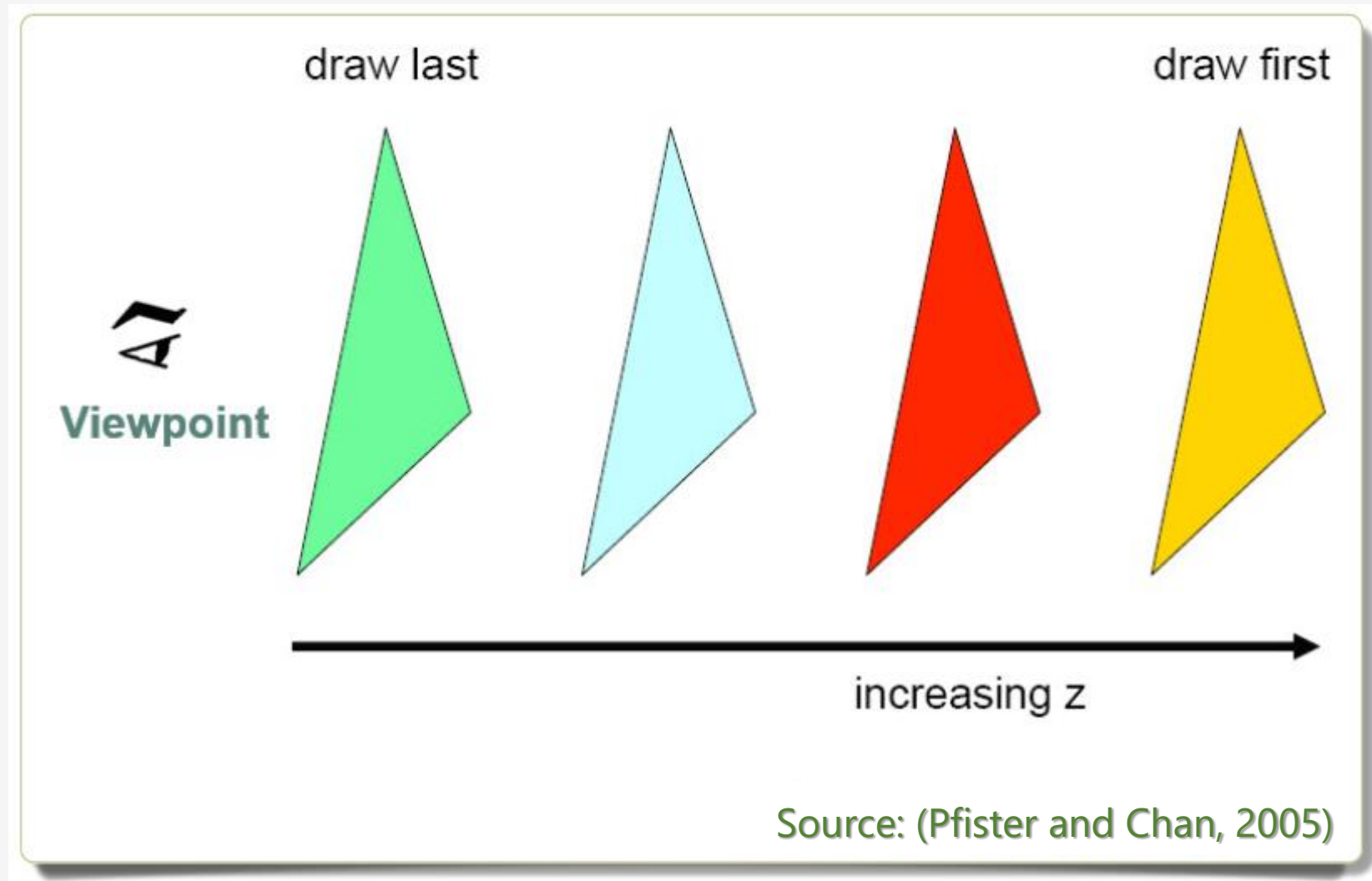
Using Double Buffering

- ❑ By default, GLFW creates a window with a double-buffered color buffer
- ❑ You can also enable or disable it by using
`glfwWindowHint(GLFW_DOUBLEBUFFER, flag)`
where flag is Boolean
- ❑ A process of rendering a scene:
 1. Clear color buffer
`glClear( GL_COLOR_BUFFER_BIT )`
 2. Render scene
 3. Request swap of front and back buffers
`glfwSwapBuffers(window)`
 4. Repeat steps 1 - 3 for animation

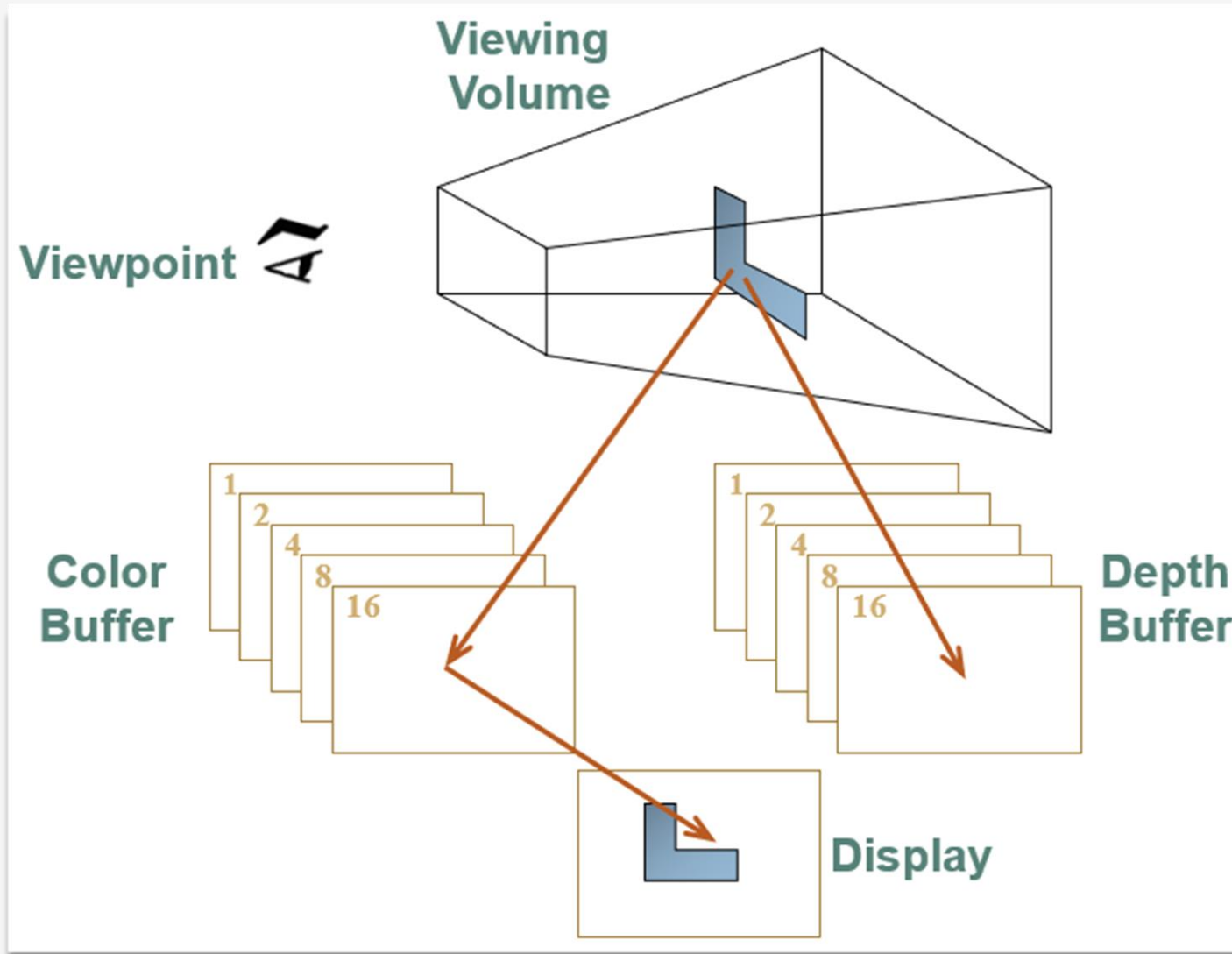
Depth Buffering and Hidden Surface Removal



Depth Buffering



Depth Buffering



Depth Buffering

- ❑ GLFW will typically create a window with a depth buffer if the specified depth bits are greater than 0. The default value is 24
- ❑ `glfwWindowHint(GLFW_DEPTH_BITS, 24)` will set depth bits to 24
- ❑ A process of rendering a scene:
 1. Enable depth buffering in OpenGL
`glEnable( GL_DEPTH_TEST )`
 2. Clear color and depth buffers
`glClear( GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT )`
 3. Render a scene
 4. Swap color buffers

References

- Dave Shreiner, Bill The Khronos OpenGL ARB Working Group, et al. OpenGL Programming Guide: The Official Guide To Learning OpenGL, Versions 3.0 and 3.1. Pearson Education, 2009.
- Dave Shreiner and Edward Angel. Interactive Computer Graphics: A top-down approach with OpenGL, 2nd Edition, 2009.
- Fletcher, M. *PyOpenGL (The Python OpenGL® Binding)*. Available at: <http://pyopengl.sourceforge.net>, accessed Oct 14, 2011.
- GLFW Community. *GLFW Documentation*. <https://www.glfw.org/documentation>, accessed May 14, 2024.